Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used proper Python indentation and HTML formatting.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, dataset, and model files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like annual_income, family_members, prediction are self-explanatory.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as home(), predict(), and model_training() are meaningful.
5	Code Comments	Inline and block comments explain logic	Yes	Comments added for preprocessing, training, and prediction sections.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate files for model training, dataset generation, and Flask application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed unnecessary imports and duplicate code.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation is present; advanced exception handling can be improved.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python, Pandas	Model training and future datasets	High
2	Machine Learning Model	Scikit-learn	Prediction system and deployment	High
3	Flask Prediction API	Flask	Web application and future ML projects	High
4	HTML Form Templates	HTML, CSS	User input and result pages	Medium
5	Data Visualization Scripts	Matplotlib, Seaborn	Dataset analysis and reporting	Medium

Overall Code Quality Assessment:

Aspect	**Rating (1-5)**	**Comments**
Code Layout & Structure	5	Well-organized project folders and files.
Readability	5	Clear variable names and proper formatting.
Reusability	5	Modules can be reused in similar prediction projects.
Documentation / Comments	4	Adequate comments provided; more detailed documentation can be added.
Overall Score	4.75/5	The project follows good coding practices and is easy to maintain and extend.